CROSS-EF-05-LEMMA-bound v1: W vs L3

ZFL Lab

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Define $W(t) := \int_{R^3} \omega^T S\omega \, dx$.
Identity (div-free u, ω): $W = -\int u_i(\partial_j \omega_i)\omega_j \, dx$.
Holder: $|W| \le \|u\|_{L^3} \|\nabla\omega\|_{L^2} \|\omega\|_{L^6}$.
Sobolev in R^3: $\|\omega\|_{L^6} \le C\|\nabla\omega\|_{L^2}$.
Therefore:

$$\boxed{|W(t)| \le C\|u(t)\|_{L^3} \|\nabla\omega(t)\|_{L^2}^2.}$$

Scale audit: $u_\lambda(x,t) = \lambda u(\lambda x, \lambda^2 t)$, $\|u_\lambda\|_3 = \|u\|_3$, $\|\omega_\lambda\|_2 = \lambda^{1/2}\|\omega\|_2$, $W_\lambda = \lambda^3 W$. Proposed bound $|W| \lesssim \|u\|_3^a \|\omega\|_2^b$ would need $b = 6$ homogeneous, not available.

Consequence:

If $\sup_{t<T_*} \|u(t)\|_3 = M < \infty$ then $|W(t)| \le CM\|\nabla\omega(t)\|_2^2$.

This does NOT imply integrability of W nor boundedness in time, nor eliminates dissipation in enstrophy identity of E-04.

Classification: $\boxed{\text{RELATION PROVEN: } |W| \lesssim \|u\|_3 \|\nabla\omega\|_2^2}$ $\boxed{C_E^{weak} \iff C_{L^3} \text{ NOT PROVEN, OPEN}}$

Hint: If quantitative bound on $\|u\|_3$ were obtained, interaction with dissipation from E-04 could give bridge.

State remains $\mathcal{C}_{PROVEN} = \{C_E^{weak}, C_{L^3}\}$ in main.